higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(First Paper)
NQF LEVEL 4

21 FEBRUARY 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD	Reading from a table/Reading from a graph/drawing
F	Choosing correct formula
SF	Substitution in formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off
MF	Manipulating formula

This marking guideline consists of 9 pages.

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QUESTION 1 *Do NOT deduct marks if the R or % sign are omitted.		
Question	**Solution**	**Explanation**
1.1	$\frac{\sqrt{36}+\sqrt[3]{8}+2^3}{3(16+2)-22}$ $= \frac{6+2+8\checkmark}{54-22\checkmark}$ $= \frac{16}{32}\checkmark$ $= \frac{1}{2}\checkmark$	1 numerator 1 denominator 1 S 1A (1 answer only) (4)
1.2	-0,2; 0,2;✓ 0,201; 0,23✓	2 A one mark for every 2 decimals in correct order (2)
1.3	$\frac{500\,000}{1000}$ ✓ = 500✓ kg $\therefore \frac{500\text{ kg}}{1000}$ = ✓0,5✓ tonnes OR $\frac{500\,000}{1000\,000}$ ✓✓✓ = 0,5✓ tonnes	1 M 1 C 500 1 C ÷ 1 000 1 A 0,5 tonnes 2 C 1 000 000 1 M 500 000 ÷ 1 000 000 1 A 0,5 tonnes (4)
1.4	351✓; 1 053✓; 3 159✓	3 A (3)
1.5	Price of shoe = $\frac{1\,296\checkmark \times 100\checkmark}{108\checkmark}$ = R1 200✓	3 MA 1 A (4)
1.6	Time in Singapore = 16:58 + 6 hours✓ = 22:58✓ **OR** 10:58 pm✓✓	1 M 1 A 2 MA (2 answers only) (2)

1.7	Petrol consumption $= \frac{423\text{ km}}{39,7\text{ litres}}$ ✓ $= 10,65$✓ $= 10,7$✓ km/ℓ	1 M 1 A 1 R (3 answer only) (3)
1.8	$\frac{229}{60}$ = 3 hours✓ + 49 minutes✓	2 MA (2 answer only) (2)
1.9	$\text{Time} = \frac{\text{distance}}{\text{speed}}$✓ $\text{Time} = \frac{115\text{ km}}{127\text{ km/ h}}$✓ = 0,91 hours✓	1 MF 1 SF 1 A (3)
1.10	300 : 500 : 200 3✓: 5✓ : 2✓	3 A (3)

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QUESTION 2 *Penalise once for incorrect unit.			
Question	**Solution**		**Explanation**
2.1	2.1.1	3✓	1 A (1)
	2.1.2	Total area = 6 × 5,2✓+ 6 × 2,5✓ = 46,2✓ m^2✓	2 MA 1 A – 46,2 1 U – m^2 (4)
	2.1.3	Length = 5,2 + 2,5✓ = 7,7 m✓	1 RD 1 MA (2)
2.2	1 : 500 000 15 mm : Actual distance in mm ∴ Actual distance in mm = 15 × 500 000✓ = 7 500 000✓ Actual distance in km = 7 500 000 mm ÷ 1 000 000✓ = 7,5 km✓		1M 1A 1M 1A (4)

2.3	2.3.1	Volume = $side^2 \times height$ Side = $\sqrt{\frac{\text{volume}}{\text{height}}}$✓ $= \sqrt{\frac{3\,456✓}{24✓}}$ $= \sqrt{144}$✓ $= 12$✓cm	1 MF 2 SF 1 A 1 A (5)
	2.3.2	Radius = 12 ÷ 2✓ = 6 cm✓ (answer only full marks)	1 M 1 A (2)
	2.3.3	Volume of cylindrical container $= \pi \times r^2 \times h$ $=3.14$✓$\times 6^2$✓$\times 24$✓ $= 2\,712,96$✓ cm^3 Difference in volume = 3 456 – 2 712,96✓ = 743,04 cm^3✓	3 SF 1 A 1 M 1 CA (6)
	2.3.4	Surface area = $\pi \times diameter \times h + \pi \times r^2$ where $\pi = 3,14$ Surface area = $3,14 \times 12 \times 24$✓$+3,14 \times 6^2$✓ $=1017,36$✓cm^2✓	2 SF 1 A 1 U (4)
	2.3.5	grams✓✓	2 A (2)

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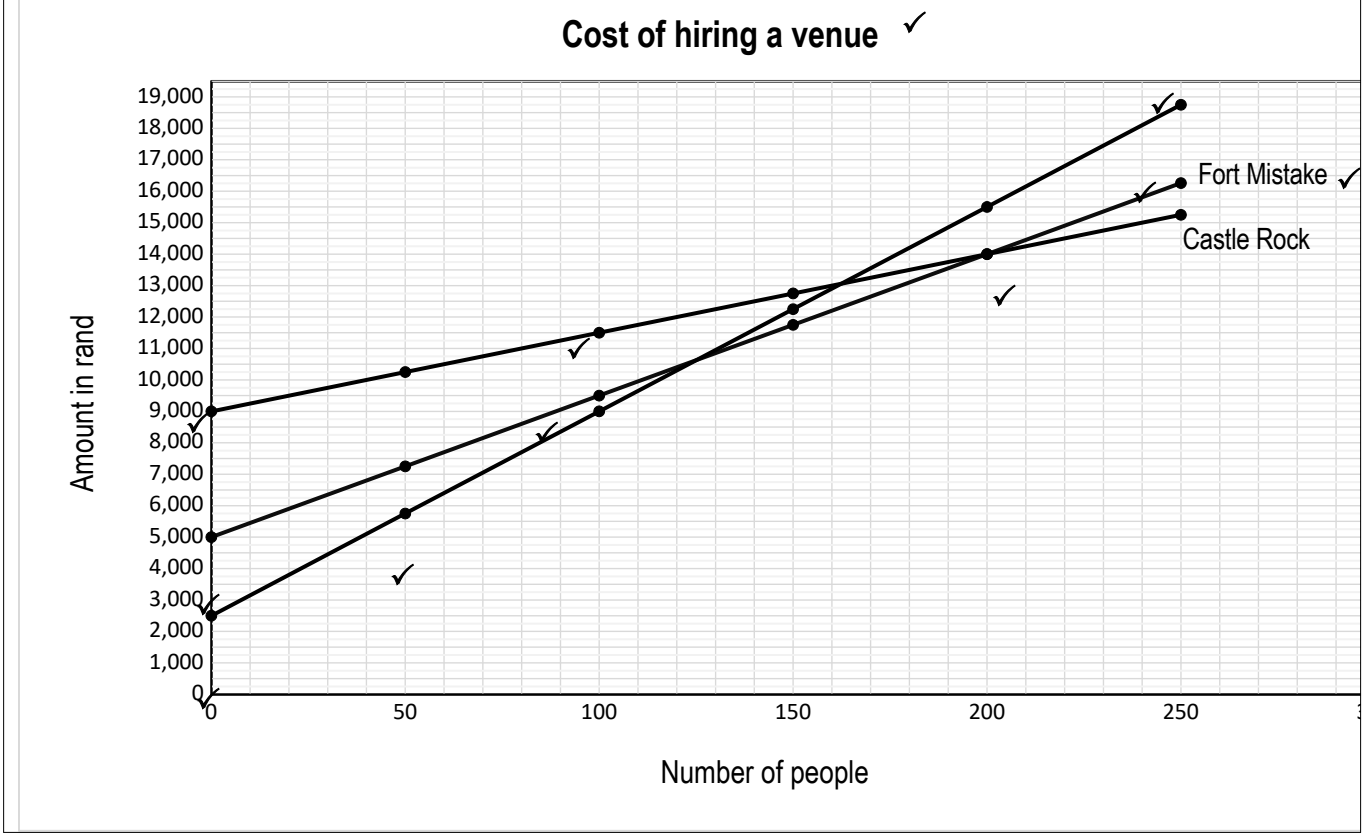
QUESTION 3 *Do NOT penalise if R sign is omitted.			
Question	**Solution**		**Explanation**
3.1	3.1.1	D✓	1 A (1)
	3.1.2	C✓	1 A (1)
	3.1.3	A✓	1 A (1)
	3.1.4	E✓	1 A (1)
	3.1.5	B✓	1 A (1)
3.2	3.2.1	R110✓	1 RT (1)
	3.2.2	Bank fees inside branch = R55 + R1,75 × 8✓ = R69✓ (answer only full marks)	1 M 1 A (2)
	3.2.3	Bank fees other ATM = R12 + R1,55 × 7✓ = R22,85✓ (answer only full marks)	1 M 1 A (2)
	3.2.4	Bank charges applicable = R6 350 – R4 500 = R1 850✓ Bank charges = R1,55✓ × 18,5✓ = R28,68✓	1 A - R1 850 2 MA 1 A (4)
	3.2.5	Bank fees for cash = R60 + R1,75 × 10 = R77,50✓ Bank fees for cheque = R55 + R7 × 1 = R62✓ Total bank charges = R77,50 + R62✓ = R139,50✓	1 MA 1 MA 1 M 1 A (4)

3.3	3.3.1	Interest is compounded✓	1 R/J (1)
	3.3.2	**A** = R30 022,47 + R4 353,26✓ = R34 375,73✓ **B** = R34 375,73 × 14,5%✓ = R4 984,48✓ **C** = R34 375,73 + R4 984,48✓ = R39 360,21✓	2 MA 1 M 1 CA 1 MA 1 CA (6)
	3.3.3	Interest = R39 360,21 – R20 000✓ = R19 360,21✓	1 M 1 CA Question 3.3.2 (2)
	3.3.4	Simple interest = R2 900✓ × 5✓ = R14 500✓ **OR** Simple interest = R20 000 × 14,5%✓ × 5✓ = R14 500✓	1 RT 1 M 1 A 2 M 1 A (3)

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QUESTION 4 * Do NOT penalise if R sign is omitted.		
Question	**Solution**	**Explanation**
4.1	A = R2 500✓ B = R16 250✓ C = R11 500✓	1 RT 1 A 1 A (3)
4.2	Total cost Fort Mistake = R5 000✓+ R45 ✓× number of people✓ **OR** Cost = 5 000✓ + 45✓*x*✓ (accept other correct alternatives)	1A – R5 000 1 A – R45 1 A – × no. of people (3)
4.3	15 350 ✓= 5 000 + 45 × number of people Number of people = 10 350✓ ÷ 45✓ = 230✓	1 SF 2 M 1 CA (Q4.1.2) (4)

4.4	Golden Sands = 2 500 + 65(120) = R10 300✓ Fort Mistake = 5 000 + 45(120) = R10 400✓ Castle Rock = 9 000 + 25(120) = R12 000✓ Therefore the cheapest venue for 120 people is Golden Sands✓	1 A 1 A 1 A 1 CA (4)
4.5	See graph below.	1 labelling title, and axes (NO mark if one is omitted) 1 labelling both lines (NO mark if one is omitted) 3 line graph – Golden Sands: (2 for plotting and 1 for line) 3 line graph – Fort Mistake: (2 for plotting and 1 for line) 3 line graph – Castle Rock: (2 for plotting and 1 for line) (11)



4.6	200✓✓	2 RG (2)
4.7	Income = R180 × 200 = R36 000✓ Expenses = R14 000✓ Profit = R36 000 – R14 000 = R22 000✓	1 MA 1 RT/RG 1 CA (3)

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QUESTION 5 *Do NOT penalise if R sign is omitted.		
Question	**Solution**	**Explanation**
5.1	Inflation is the comparison on increase in the price you pay for goods✓ month to month, year on year.✓	2 E (2)
5.2	Increase = 12%✓	1 RT (1)
5.3	**A** = R45,32✓– R38,99✓ = R6,33✓	2 MA 1 A (3)
5.4	**B** = $\frac{\text{- R 7,58}✓}{\text{R 28,55}✓} \times 100$ = -26,5% ✓ Accept -27%	2 MA 1 A (3)
5.5	Ascending order: **Jan 2017** **Price** ✓ R19,30 R45,32 R61,16 R36,48 R81,16 R55,49 ✓ R91,21 R20,97 R63,60 R41,94 ✓ R80,82 R176,47 R244,32 Median = R91,21✓	 2 A – ascending order 1 M – locating the position of the median 1 A – median (4)

5.6	Range = R244,32✓ – R19,30✓ = R225,02✓		2 M/RT 1 A (3)
5.7	Mean Jan 2017 = $\frac{\text{R 1018.24}}{13}$✓ = R78,33✓ Mean Jan 2016 = $\frac{\text{R 905,75}}{13}$✓ = R69,67✓ **Difference =** R78,33 – R69,67✓ = R8,66✓		2017: 1 MA 1 A – mean 2016: 1 MA 1 A – mean 1 M 1 A (6)
5.8	Total percentage of the pie chart = 36% + 43% + 15% = 94%.✓ A pie chart should add up to 100%.✓		2 E (2)
5.9	5.9.1	Compound bar graph✓	1 A (1)
	5.9.2	Graph A✓ The prices on the vertical axis start at R56,50✓ and NOT at zero.✓ The prices on the vertical axis increases by only 50c.✓ Thus causing the change in price to look bigger than it really is.✓	1 A 2 R/J 2 R/J (5)

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TOTAL: 150

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